Visvesvaraya Technological University

Belgaum, Karnataka- 590014

Third Year

A Mini-Project Report

On

" Explainable Multimodal Brain – Heart Stress Detection and AI-Driven Intervention System using EEG and ECG Signals "

Submitted in the partial fulfilment of the requirements for the award of the Degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

Submitted by

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2025–2026

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2025–2026

Certificate

This is to certify that the Mini Project Work entitled **"Explainable Multimodal Brain – Heart Stress Detection and AI-Driven Intervention System using EEG and ECG Signals"** is a bonafide work carried out by **Ashlesh (1DS23CD012), Sambhram S Naregal (1DS23CD045), Tushar D Kudachimath (1DS23CD058) and Vishal Krishna Bhat K (1DS23CD063)** , in partial fulfilment for the VI semester of Bachelor of Engineering in CSE (Data Science) of the Visvesvaraya Technological University, Belgaum during the year 2025–2026. The Project report has been approved as it satisfies the academic requirements prescribed for the Bachelor of Engineering degree.

Signature of Project Coordinator	Signature of Guide	Signature of HOD
[Dr. Aarathi S]	[Dr. Aarathi S]	[Dr. Harish H. Kenchannavar]

Name of the Examiners	**Signature with Date**
1.	
2.	

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We extend our sincere thanks to our family members and our beloved friends, who had been strongly supporting us in all our endeavors.

CONTENTS

Chapter 1

Template Usage Guide

This chapter contains sample examples for commonly used LaTeX components in this template.

1.1 Section Example

This is a normal section example.

1.1.1 Subsection Example

This is a subsection example.

1.2 Bullet List Example

- Point One
- Point Two
- Point Three

1.3 Numbered List Example

1. First Point
2. Second Point
3. Third Point

1.4 Table Example

Table 1.1: Sample Table

Column 1	Column 2
Data A	Data B
Data C	Data D

1.5 Image Example

Figure 1.1: Sample Image

1.6 Citation Example

Add references inside `references.bib`

Example:

```
@article{r1,
author = {John Smith},
title = {Artificial Intelligence in Clinical Systems},
journal = {Journal of Medical Computing},
year = {2024}
}
```

Use citations like:

```
\cite{r1}
```

Example:

Artificial Intelligence improves healthcare systems [2].

1.7 Code Example

```
print("Hello World")
```

1.8 Equation Example

$$E = mc^2$$

1.9 Important Notes

- Put all images inside the images/ folder
- Compile multiple times if TOC or references do not update
- Add references inside references.bib

Chapter 2

Introduction

Write the complete introduction of the project here.

Chapter 3

Literature Survey

Discuss existing systems and related research.

Chapter 4

Research Gap Identified

Explain the limitations in existing systems.

Chapter 5

Architectures

5.1 High Level Architecture

Insert high level architecture diagram here.

5.2 Detailed Architecture

Insert detailed architecture diagram here.

5.3 UML Diagrams

- Use Case Diagram
- Class Diagram
- Sequence Diagram
- Activity Diagram

Chapter 6

Project Objectives

- Objective 1
- Objective 2
- Objective 3
- Objective 4

Chapter 7

Timeline using Gantt

Insert Gantt chart here.

Chapter 8

Hardware and Software Requirements

8.1 Hardware Requirements

- Processor
- RAM
- Storage

8.2 Software Requirements

- Operating System
- Programming Language
- Frameworks
- Database

Chapter 9

Sample Code

```
Example code here
```

Chapter 10

Project Demo Screenshots

Insert screenshots here.

Chapter 11

AI Tools Application

Explain AI tools used in the project.

Chapter 12

SGDs Met by Project with Proper Justification

Mention SDGs achieved by the project.

Chapter 13

Purpose of the Project

Explain target audience and future scope.

Chapter 14

Research Paper Draft

Attach research paper draft and Turnitin report. Artificial Intelligence is transforming healthcare systems rapidly [2, 1].

References and Bibliography

[1] Jane Doe. Predictive analytics in hospitals. *International Healthcare Review*, 2023.

[2] John Smith. Artificial intelligence in clinical systems. *Journal of Medical Computing*, 2024.